

AI SMART CITY

COMPLAINT MANAGEMENT & PREDICTIVE ANALYTICS PLATFORM

PROJECT DOCUMENTATION

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GitHub	https://github.com/chinni02/AI-Smart-City

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to everyone who supported the design, development, testing, and documentation of the AI Smart City Complaint Management & Predictive Analytics Platform. This is an individual self-project developed to apply full-stack development, artificial intelligence, database management, and predictive analytics to a practical civic problem.

I am thankful for the official documentation, open-source technologies, developer communities, and learning resources that supported the implementation of HTML, CSS, JavaScript, Python, FastAPI, PostgreSQL, SQLAlchemy, AI-based complaint classification, predictive analytics, and data visualization. These resources provided valuable guidance throughout the development process and helped in building an efficient platform for managing citizen complaints, analyzing complaint patterns, and supporting data-driven decision-making for smart city management.

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ABSTRACT

The AI Smart City Complaint Management & Predictive Analytics Platform is a web-based application designed to improve the management and analysis of citizen complaints related to urban services. Traditional complaint management systems mainly focus on recording complaints and tracking their status, which can make it difficult to identify priority issues and understand city-wide complaint patterns.

The proposed system combines a React-based frontend, FastAPI backend, PostgreSQL database, and AI-based analysis to provide a centralized complaint management platform. Citizens or administrators can submit complaints by providing a description and location. The system processes the complaint using AI-based classification to identify its category and priority, along with a confidence score and relevant keywords. A predictive risk analysis module also evaluates the complaint and provides a risk score, risk level, and associated factors.

The dashboard provides real-time information about total complaints, high-priority complaints, frequently reported categories, affected locations, complaint status, and priority distribution. Administrators can search, filter, update, and delete complaints through the dashboard. The system also implements authentication with protected dashboard access and logout functionality.

Overall, the project demonstrates how artificial intelligence, predictive analytics, web technologies, and database management can be combined to create a smarter and more organized approach to urban complaint management.

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1.INTRODUCTION

Smart cities use digital technologies to improve public services, infrastructure, and the quality of life of citizens. One important aspect of smart city management is the effective handling of complaints related to roads, garbage, electricity, water, traffic, drainage, and other public services.

In conventional complaint management systems, complaints are generally stored and manually processed. This approach provides limited intelligence for identifying the type and urgency of a complaint. It can also make it difficult for administrators to understand complaint trends and determine which issues require immediate attention.

The **AI Smart City Complaint Management & Predictive Analytics Platform** addresses these challenges by integrating complaint management with artificial intelligence and predictive analytics.

The system provides a web-based dashboard where complaints can be submitted and managed. The AI classification module analyzes complaint descriptions and identifies relevant categories and priorities. The predictive risk module evaluates potential risk associated with complaints. The dashboard presents complaint information through statistics and visualizations, helping administrators understand city-wide issues.

The system is developed using React for the frontend, FastAPI for the backend, PostgreSQL for database management, and Python-based AI modules for intelligent analysis.

2.PROBLEM STATEMENT

Traditional urban complaint management systems often depend heavily on manual processing. When the number of complaints increases, it becomes difficult to classify complaints, identify high-priority issues, track their status, and analyze overall complaint patterns efficiently.

The major problems include:

- Manual classification of complaints.
- Difficulty in identifying high-priority complaints.
- Lack of predictive analysis.
- Limited visualization of complaint trends.
- Difficulty in searching and filtering complaints.
- Lack of centralized complaint management.
- Difficulty in monitoring complaint status.
- Limited support for data-driven decision making.

Therefore, there is a need for an intelligent complaint management platform that can automatically analyze complaints, identify their category and priority, provide predictive risk information, and present useful city-level analytics.

3. OBJECTIVES

The main objective of the project is to develop an intelligent platform that simplifies city complaint management and provides useful analytical insights to administrators.

The major objectives are:

3.1 Complaint Management

To provide a centralized platform for creating, viewing, updating, and deleting city-related complaints.

3.2 AI-Based Complaint Classification

To use artificial intelligence to analyze complaint descriptions and assist in identifying the appropriate complaint category.

3.3 Complaint Prioritization

To assist administrators in identifying high-priority complaints that require immediate attention.

3.4 Risk Prediction

To provide a risk prediction component that can analyze complaint-related information and generate a complaint risk prediction.

3.5 City Analytics

To provide city-level analytical information based on the collected complaints.

3.6 Data Visualization

To represent complaint information through graphical dashboards, making important statistics easier to understand.

3.7 Status Monitoring

To enable administrators to monitor complaints according to their current status, such as pending, in progress, and resolved.

3.8 Location-Based Analysis

To identify locations with a higher concentration of reported complaints and provide administrators with useful geographical insights.

3.9 Secure Access

To provide authentication functionality so that authorized users can access the administrative features of the platform.

3.10 Improve Decision-Making

To support administrators in making better decisions by combining complaint information, AI analysis, risk prediction, and visual analytics in one platform.

4.EXISTING SYSTEM

In a traditional complaint management system, citizens report problems through manual forms, phone calls, websites, or basic online portals. The complaints are generally stored in a database and later processed by administrators.

Limitations of the Existing System

- Complaint classification may require manual effort.
- Priority assignment may depend on human judgment.
- There is limited predictive analysis.
- Complaint trends may not be easily visible.
- Administrators may need to manually search through complaints.
- Decision making is not strongly supported by data analytics.
- Large numbers of complaints can become difficult to manage.

Traditional systems are therefore mainly focused on recording and tracking complaints, rather than intelligent analysis and prediction.

5.PROPOSED SYSTEM

The proposed **AI Smart City Complaint Management & Predictive Analytics Platform** provides an intelligent solution for managing urban complaints.

The system allows authorized users to log in and access a dashboard. A complaint can be submitted by entering its description and location. After submission, the system sends the information to the backend and performs AI-based analysis.

The system provides:

- Complaint management.
- AI-based classification.
- Priority prediction.
- Confidence measurement.
- Keyword identification.
- Predictive risk analysis.
- Risk score and risk level.
- City-level analytics.
- Category visualization.
- Status visualization.
- Priority visualization.
- Search and filtering.
- Complaint updating.
- Complaint deletion.
- Authentication and protected routes.

The proposed system therefore combines complaint management + artificial intelligence + predictive analytics + visualization into one platform.

6. TECHNOLOGIES USED & MODULES

6.1 Technologies Used

The AI Smart City Complaint Management and Predictive Analytics Platform is developed using modern frontend and backend technologies. The system follows a client-server architecture in which the frontend communicates with backend REST APIs.

6.1.1 Frontend Technologies

React:

React is used to develop the interactive user interface of the application. The frontend contains separate pages for login and the main dashboard.

Vite:

Vite is used as the frontend development and build tool. It provides a fast development environment for the React application.

JavaScript and JSX:

JavaScript is used for application logic, while JSX is used to define React components and user interface elements.

CSS:

CSS is used to design and style the application interface, including forms, dashboards, cards, charts, buttons, filters, and other visual components.

Axios

Used to communicate between the React frontend and FastAPI backend.

Recharts

Used to create graphical representations such as bar charts and pie charts.

6.1.2 Backend Technologies

Python

Used as the primary backend and AI programming language.

FastAPI

Used to develop REST API endpoints for authentication, complaint management, AI classification, risk prediction, and analytics.

Database

PostgreSQL

Used to store complaint and related application data.

SQLAlchemy

Used as the ORM for interacting with the PostgreSQL database.

Authentication

The system uses token-based authentication to control access to the dashboard. The access token is stored in the browser and used to maintain the login session.

Development Tools

- Visual Studio Code
- Git
- GitHub
- PowerShell
- Web Browser

6.2 Project Modules

The major modules implemented in the project are as follows.

6.2.1 Authentication Module

The Authentication Module provides secure access to the application.

Provides:

- Login.
- Access token generation.
- Protected dashboard access.
- Logout.
- Redirection to login when authentication is unavailable.

6.2.2 Complaint Management Module

The Complaint Management Module is responsible for handling city-related complaints.

The module provides functionality to:

- Create complaint.
- View complaints.
- Update complaint.
- Delete complaint.
- Search complaints.
- Filter complaints.

6.2.3 AI Classification Module

Analyzes the complaint description and identifies:

- Category.
- Priority.
- Confidence score.
- Matched keywords.

6.2.4 Predictive Risk Analysis Module

Evaluates complaint information and provides:

- Risk score.
- Risk level.
- Risk factors.
- Critical keywords.

6.2.5 City Analytics Module

The City Analytics Module analyzes the collected complaint information and provides city-level insights.

The /analytics/city endpoint is used to retrieve city analytics.

The dashboard presents these insights through:

- Total complaints.
- High-priority complaint count.
- Most reported category.
- Most affected location.

6.2.6 6. Visualization Module

Displays complaint data using:

- Category bar chart.
- Status pie chart.
- Priority bar chart.
- Summary statistics

6.2.6 7. Dashboard Module

Provides a centralized interface for viewing and managing all major project functionalities.

7.REQUIREMENT ANALYSIS:

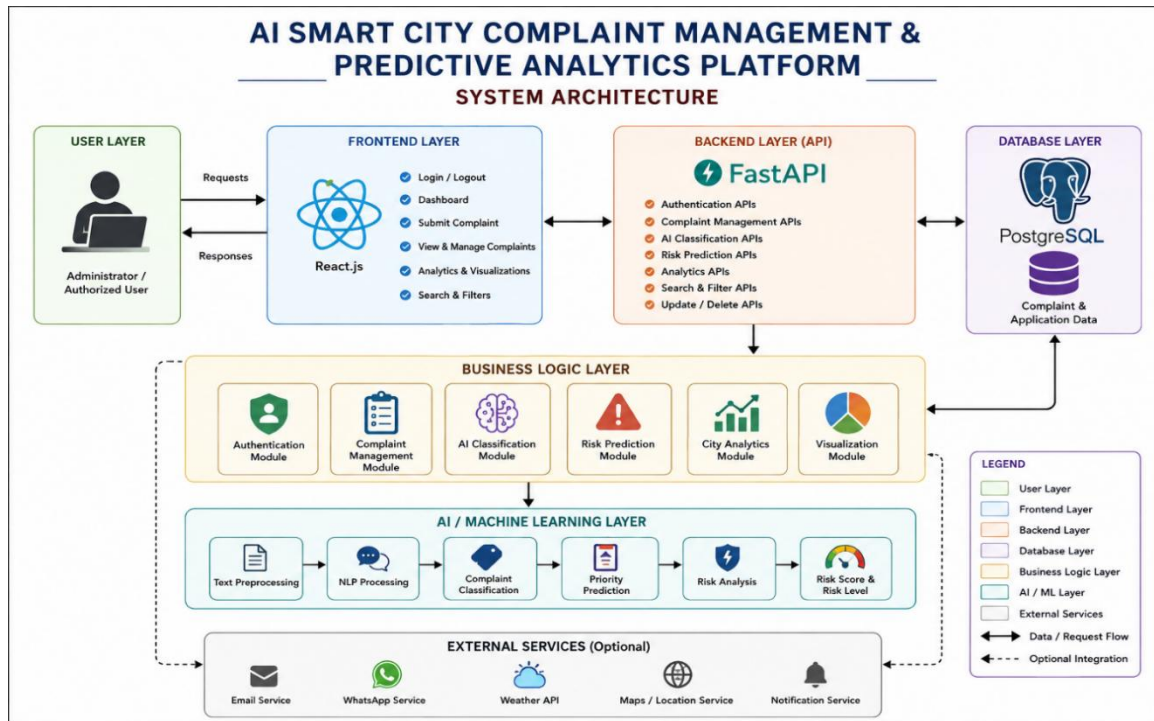
HARDWARE REQUIREMENTS:

- System : Intel i3 Processor
- Hard Disk : 1 TB
- Monitor : 14' Colour Monitor
- Mouse : Optical Mouse
- Ram : 4GB

SOFTWARE REQUIREMENTS:

- Operating system : Windows 11
- Coding Language : Python
- Front-End : React.js
- Back-End : FastAPI
- ORM : SQLAlchemy
- Data Base : PostgreSQL

8. SYSTEM ARCHITECTURE



1. User/Admin Layer:

The administrator interacts with the system through the web dashboard to submit, view, update, filter, and delete complaints.

2. Frontend Layer – React.js:

The React.js frontend provides the user interface for complaint submission, login, dashboard visualization, AI results, charts, and complaint management.

3. Backend Layer – FastAPI:

FastAPI handles API requests from the frontend, manages authentication, complaint operations, AI processing, predictive analysis, and communication with the database.

4. AI & Analytics Layer:

This layer analyzes complaints using AI-based classification and predictive risk analysis. It identifies the complaint category, priority, keywords, confidence, and risk level. It also generates city-level analytics.

5. Database Layer – PostgreSQL:

PostgreSQL stores user authentication information and complaint-related data such as description, location, category, priority, status, and analysis results.

6. Output/Visualization Layer:

The processed information is displayed on the dashboard using statistics, charts, complaint lists, AI classification results, and predictive risk analysis.

9. IMPLEMENTATION

9.1 User Authentication:

The user enters a username and password on the login page. The frontend sends the credentials to the FastAPI authentication endpoint.

After successful authentication, an access token is received and stored in local storage. The user is then redirected to the dashboard.

Protected routing prevents unauthorized users from directly accessing the dashboard.

When the user logs out, the stored access token is removed and the application redirects the user back to the login page.

9.2 Complaint Submission:

The dashboard contains a complaint submission form with:

- Complaint Description
- Location

After entering the required information, the user clicks Submit & Analyze.

The frontend sends the complaint to the FastAPI backend.

9.3 AI Classification:

After a complaint is submitted, the AI classification endpoint analyzes the complaint.

The system returns information such as:

- Complaint category.
- Priority.
- Confidence.
- Matched keywords.

The results are displayed on the dashboard in an AI Classification card.

9.4 Predictive Risk Analysis:

The submitted complaint is also processed by the risk prediction module.

The module provides:

- Risk score.
- Risk level.
- Risk factors.
- Critical keywords.

This allows administrators to identify complaints that may require greater attention.

9.5 Complaint Database Management:

Complaint records are stored in PostgreSQL.

The application supports:

- Creating complaints.
- Retrieving complaints.
- Updating complaints.
- Deleting complaints.

The React dashboard communicates with the backend using API requests.

9.6 City Intelligence:

The City Intelligence section provides an overall view of complaint activity.

It displays:

- Total complaints.
- High-priority complaints.
- Most reported category.
- Most affected location.

This provides administrators with a quick overview of current complaint conditions.

9.7 Data Visualization:

The dashboard uses Recharts to visualize complaint information.

Three major visualizations are provided:

1. Complaints by Category
2. Complaints by Status
3. Complaints by Priority

These visualizations make it easier to understand complaint distribution and urgency.

9.8 Complaint Search and Filtering:

The dashboard allows administrators to search complaints using:

- Complaint description.
- Location.

Complaints can also be filtered based on:

- Category.
- Priority.
- Status.

A Clear Filters option resets all selected filters.

9.9 Complaint Updating:

Administrators can update:

- Complaint priority.
- Complaint status.

The system sends the update to the backend and refreshes the complaint list.

9.10 Complaint Deletion:

Administrators can delete unwanted complaint records.

Before deletion, the system asks for confirmation. After successful deletion, the complaint list and city analytics are refreshed.

10. OUTPUT SCREENSHOTS

Figure 10.1 — Login Page

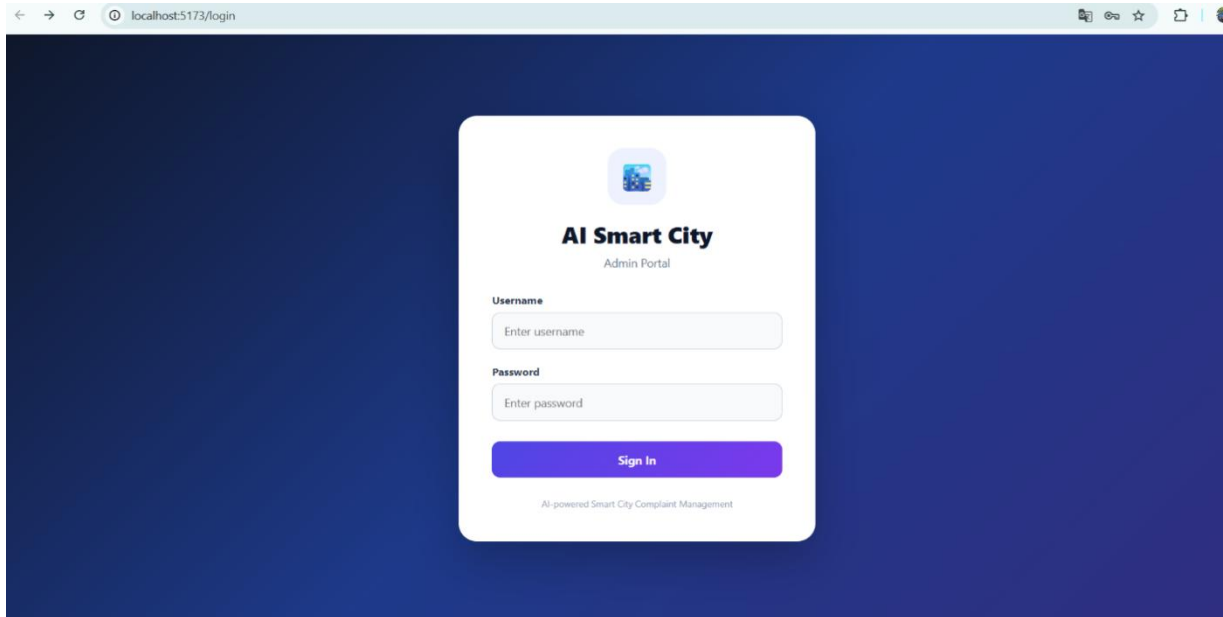


Figure 10.2 — Dashboard - Complaint Submission

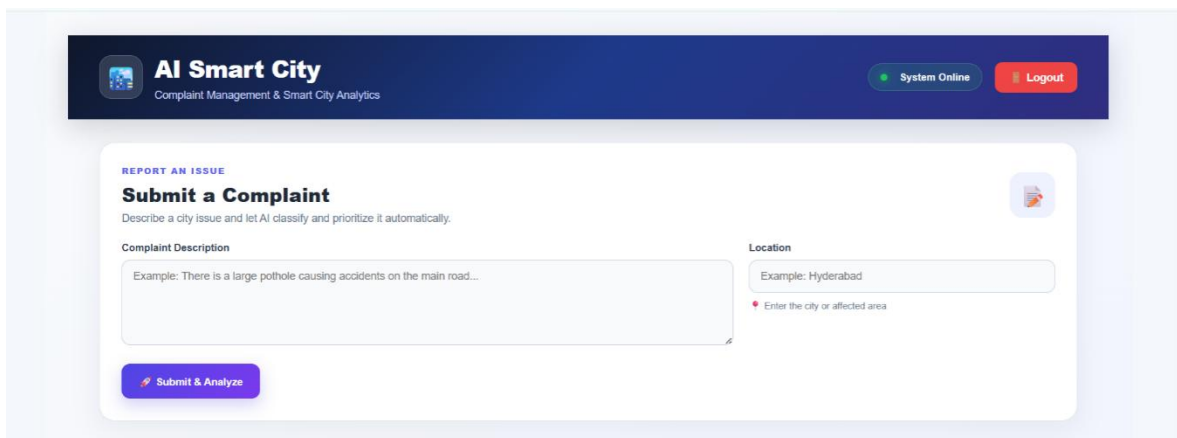


Figure 10.3 — City Intelligence

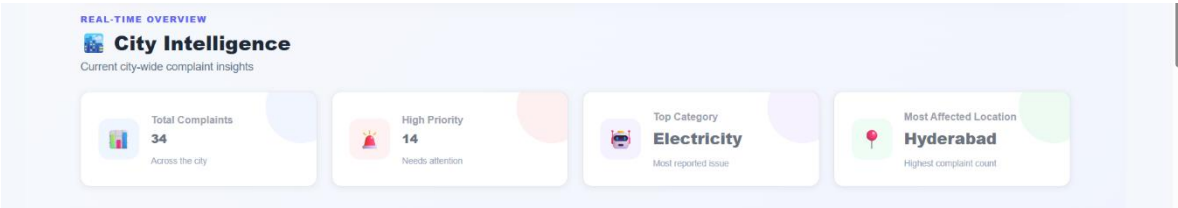


Figure 10.4 — Smart City Analytics

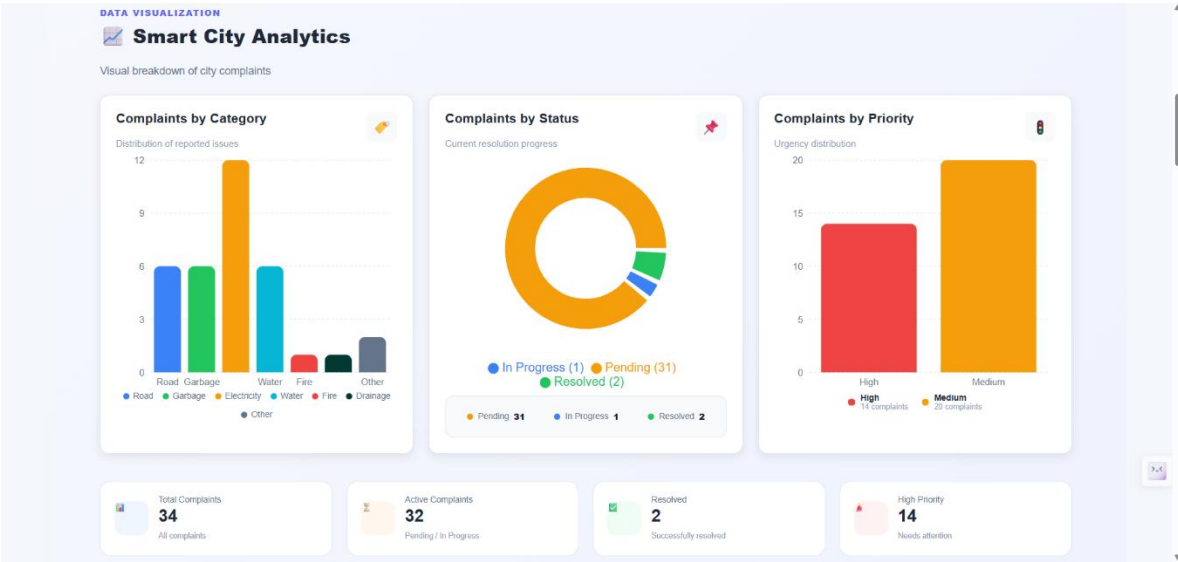


Figure 10.5 — Search and Filter

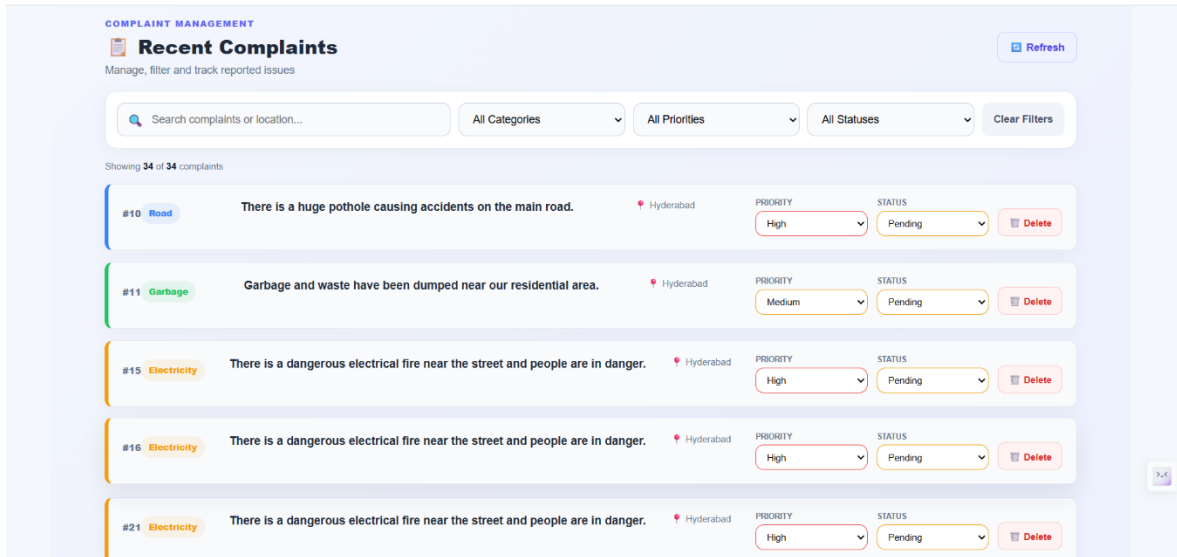
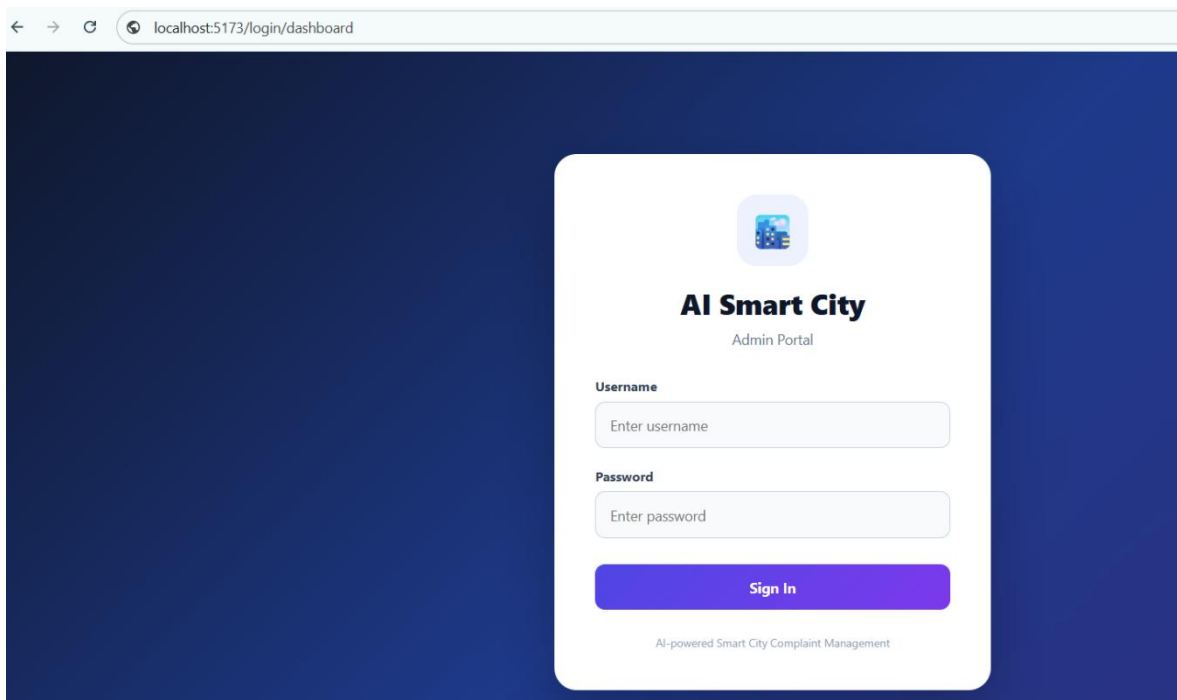


Figure 10.6 — Protected Dashboard



11. ADVANTAGES

The major advantages of the proposed system are:

1. Centralized complaint management.
2. AI-based complaint classification.
3. Automated priority identification.
4. Predictive risk assessment.
5. Confidence-based AI results.
6. Easy complaint search and filtering.
7. Real-time dashboard statistics.
8. Visual representation of complaint data.
9. Easy complaint status management.
10. Protected dashboard access.
11. Reduced manual effort.
12. Better support for data-driven decision making.
13. User-friendly interface.
14. Scalable architecture for future improvements.

12. LIMITATIONS

Although the system provides several useful features, it has some limitations:

1. AI classification depends on the quality of the complaint description.
2. The current system has limited training data.
3. Risk prediction may not represent real-world emergency conditions perfectly.
4. The system currently focuses mainly on complaint-based analysis.
5. Advanced geographic mapping is not currently implemented.
6. Real-time external city sensor data is not integrated.
7. The system requires a properly configured backend and PostgreSQL database.
8. The current platform is primarily designed as an administrative complaint management system.

13. APPLICATIONS

The system can be applied in various smart city scenarios, including:

Municipal Complaint Management:

Municipal authorities can manage complaints related to public services.

Road Maintenance:

Road and pothole complaints can be classified and prioritized.

Waste Management:

Garbage-related complaints can be monitored and analyzed.

Electricity Services:

Electricity-related issues can be recorded and prioritized.

Water Management:

Water supply and related complaints can be tracked.

Traffic Management:

Traffic-related complaints can be analyzed for better planning.

Drainage Management:

Drainage-related issues can be identified and monitored.

Urban Planning:

Complaint analytics can provide useful information for future infrastructure planning.

14.FUTURE SCOPE

The project can be further enhanced in several ways.

1. Advanced Machine Learning Models:

More advanced machine learning and deep learning models can be introduced to improve classification accuracy.

2. Real-Time Data Integration:

The system can be connected with IoT sensors and other smart city data sources.

3. Geographic Information System:

A map-based interface can be added to display complaints geographically.

4. Mobile Application:

A mobile application can allow citizens to submit complaints directly from their smartphones.

5. Image-Based Complaint Analysis:

Users could upload photographs of issues such as potholes, garbage, or damaged infrastructure, and computer vision models could analyze them.

6. Automated Notifications:

Email, SMS, or application notifications could be added for complaint updates.

7. Advanced Prediction:

Historical complaint data could be used to predict areas where certain types of complaints are likely to increase.

8. Multilingual Support:

The platform could support multiple Indian languages to make the system more accessible.

9. Advanced AI Assistant:

An AI chatbot could be integrated to help users report complaints and obtain complaint status information.

10. Cloud Deployment:

The application could be deployed on cloud infrastructure for improved scalability and accessibility.

15.CONCLUSION

The AI Smart City Complaint Management & Predictive Analytics Platform demonstrates how artificial intelligence, web development, database management, and predictive analytics can be combined to improve urban complaint management.

The system provides a centralized dashboard through which complaints can be submitted, analyzed, searched, filtered, updated, and deleted. The AI classification module helps identify complaint categories and priorities, while the predictive risk module provides additional information about potential risks.

The dashboard also provides city-level statistics and visualizations that help administrators understand complaint patterns. Authentication and protected routes provide controlled access to the management system.

Overall, the project provides a practical foundation for intelligent smart city complaint management and can be further extended with advanced machine learning, computer vision, GIS, IoT, mobile applications, and cloud deployment.

16.GITHUB

GitHub Repository:

<https://github.com/chinni02/AI-Smart-City>

17. REFERENCES

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